

TIC 65970288

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.370534 d
Epoch (t0) 1492.9125 BTJD
Duration 0.0840 d
Depth 1.169 % (11,694 ppm)
SNR 52.8

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 16 (gate ≥ 3)
Robust depth ratio 0.97 (gate ≥ 0.50)
Robust depth sigma 36.8 (gate ≥ 7.0)
Odd/even z-score 0.18 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

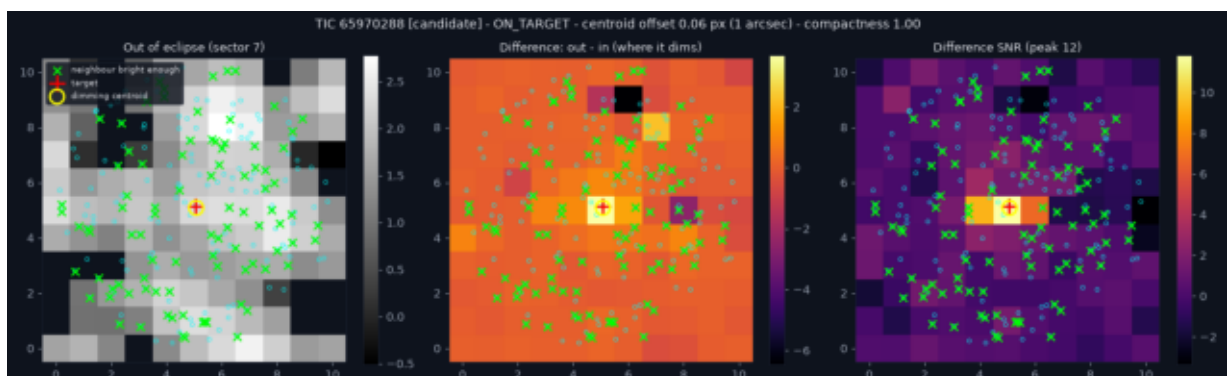
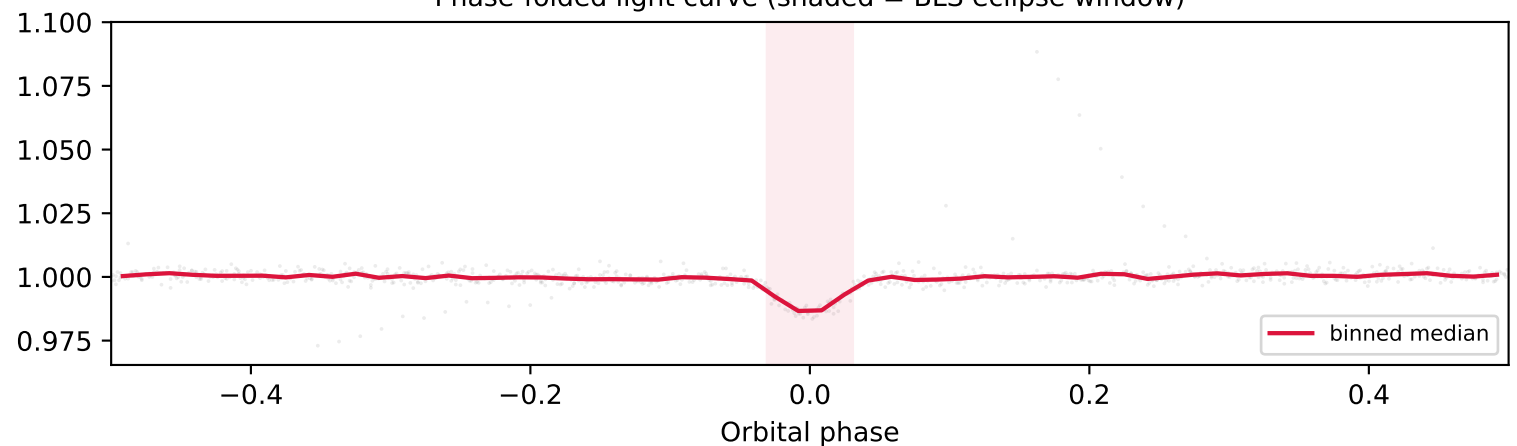
VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 7 light-curve dip in this sector: 35.9 sigma t0 refined by 0.0050 phase
Difference-image peak SNR 11.82 (gate ≥ 5.0) Compactness 1.00 (gate ≥ 0.50)
Dimming centroid offset 0.06 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 65970307
0.91 px away, dTmag +3.34 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.